Graphs of Polynomial Functions (2.2) Name ______________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write "2" in the blank. Solve that problem, find your answer and write "3" in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 2 # ______ Describe the right-hand end behavior of $f(x) = 2x^3$.	ANSWER: 5 # ______ Describe the end behavior (both ends) of $f(x) = -x^4 + 2$.
ANSWER: 6 # ______ State the multiplicity of the zero at $x = 2$ in $f(x) = (x-2)^3(x+1)$.	ANSWER: $(0, 6)$ # ______ State the maximum number of turning points of a degree-2 polynomial.

Answer: up

_______ State the maximum number of turning points of a degree-7 polynomial.

Answer: cross

_______ State the y-intercept of $f(x) = (x-1)(x+2)(x-3)$.

Answer: bounce

_______ State the maximum number of real zeros of $f(x) = x^5 - x$.

Answer: 1

_______ How many distinct real zeros does $f(x) = (x+1)^2(x-4)$ have?

Answer: 3

_______ At a zero of *even* multiplicity, does the graph cross or bounce?

Answer: down

_______ At a zero of *odd* multiplicity, does the graph cross or bounce?